One-Day Value at Risk for a Multi-Asset EUR Portfolio

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1 The Core Idea

Value at Risk (VaR) answers one question: how much can the portfolio lose in one day, with 99% confidence? Expected shortfall (ES) adds a second: if the loss does exceed the VaR, how large is it on average? A risk manager needs both, because a limit is set on the first number and capital is consumed by the second.

The repository applies four VaR models to one portfolio and tests them out of sample. The portfolio is held by a EUR investor: four Paris stocks, three US stocks, a US Treasury ETF (TLT) and Ethereum, rebalanced daily to fixed weights.

- **Why compare models:** the choice of distribution moves the 99% VaR by about 0.4 percentage points and the ES by more than 1 point on the same data.
- **Why backtest:** a model that fits the full sample can still fail day by day, and a VaR that is never checked against realised losses is only an opinion.

2 Why It Matters

- **The tail, not the average, sets the limit.** Over 1,494 days the portfolio earned a mean of 0.098% a day with a daily volatility of 1.275%. The 1% tail lies more than two and a half volatilities below that mean, and that is where limits and capital are decided.
- **VaR hides the size of the loss behind it.** The historical and the Student-t models give the same VaR (3.37%) but different ES (4.77% against 4.63%). The normal model has an ES only 1.15 times its VaR, the historical data 1.42 times. Two models that agree on one number can disagree on the cost of a bad day.
- **Losses are asymmetric.** After a loss d, the gain needed to recover is $d/(1-d)$. The portfolio lost 11.06% on 12 March 2020, which needs a 12.4% gain to repair.
- **Volatility moves, and the VaR must follow.** The rolling 250-day volatility of the portfolio ranged from 12.6% to 28.0% a year, and the rolling normal VaR from 1.85% to 4.11%. A fixed number carried through that range is either too loose or too tight most of the time.

3 Assumptions

The code makes the following choices. Each line states what moves if the choice is wrong.

- **Data window.** Yahoo Finance adjusted closes from January 2020 to December 2025, which hold the March 2020 crash as well as the stress of 2022 and April 2025. In a calmer sample the historical ES would be lower: removing 2020 takes it from 4.77% to 3.70%.
- **Common dates only.** A day is kept only if all nine assets have a close (1,494 returns). Ethereum trades on weekends and Paris and New York have different holidays, so some returns span two or three calendar days. Those returns are larger than one-day returns and push the tail up slightly.
- **Simple returns, fixed weights.** $r_t = \sum_i w_i r_{i,t}$ with daily rebalancing and no costs. Without rebalancing the weights would drift towards the winners (Ethereum, NVIDIA) and the risk would rise.
- **FX and time zones.** USD assets are converted at the same-day EURUSD close, forward-filled when missing. Paris, New York and the FX quote do not close at the same time, so daily correlations across regions are probably measured too low and diversification is slightly overstated.
- **Zero mean.** The normal, Student-t and Monte Carlo models set the mean to zero; the historical estimator keeps the sample drift. With a realised drift of +0.098% a day, the parametric VaR is about 0.1 point higher than with the drift, a small conservative bias at one day and a larger one at ten.
- **Stationary, independent returns.** Each model treats the returns in its window as draws from one fixed distribution. Volatility clustering breaks this. The Christoffersen (1998) test below measures the damage.
- **Symmetric fat tails.** The Student-t fit has one degrees-of-freedom parameter ν and no skew. The portfolio skew is -0.42, so the left tail is probably a little heavier than the fitted one.
- **Monte Carlo shares one ν.** All nine assets use the portfolio's ν and one common chi-square shock. Real assets have different tails, and Ethereum is far more volatile than TLT.
- **Calibration window.** The backtest uses the previous 250 days. In such a window the 1% historical quantile rests on about 2.5 observations: it sits at position $0.01 \times 249 = 2.49$ and interpolates between the 3rd and 4th worst days. A longer window is smoother but reacts later.
- **Backtest tests.** Both tests use the χ^2_1 approximation with about 12 expected breaches, so their p-values are approximate.

4 Method

4.1 Data and portfolio

Weights: 10% each in LVMH, Sanofi, L'Oréal, Airbus, Apple, Microsoft, NVIDIA and Ethereum, and 20% in TLT. The full-sample portfolio has an annualised volatility of 20.2%, skew of -0.42 and excess kurtosis of 6.8. VaR and ES are reported as positive losses. With confidence $\alpha = 0.99$:

4.2 Historical

$$\mathrm{VaR} = -q_{1-\alpha}(r), \qquad \mathrm{ES} = -\mathbb{E}\left[r \mid r \le q_{1-\alpha}\right],$$

where $q_{1-\alpha}$ is the empirical 1% quantile (linear interpolation, `numpy.quantile`). It has no parameters. It rises when the window contains a crash and falls once the crash leaves the window.

4.3 Normal

With $z = \Phi^{-1}(\alpha) = 2.3263$ and σ the sample standard deviation:

$$\mathrm{VaR} = z\,\sigma, \qquad \mathrm{ES} = \sigma\,\frac{\varphi(z)}{1-\alpha}.$$

Everything is driven by σ: doubling the volatility doubles both numbers. The ES is always $\varphi(z)/((1-\alpha)z) = 1.146$ times the VaR, whatever the data.

4.4 Fitted t distribution

The t distribution is fitted by maximum likelihood with a zero location, giving ν and a scale s. With $q = t_\nu^{-1}(\alpha)$:

$$\mathrm{VaR} = s\,q, \qquad \mathrm{ES} = s\,\frac{f_\nu(q)}{1-\alpha}\cdot\frac{\nu+q^2}{\nu-1}.$$

A small ν means heavy tails: the full sample gives $\nu = 4.22$ and $s = 0.926\%$. As $\nu \to \infty$ the model tends to the normal. The ES formula needs $\nu > 1$.

4.5 Monte Carlo (Student-t scenarios)

Draw 100,000 joint scenarios for the nine assets: $x = z\,\sqrt{(\nu-2)/\nu}\,/\sqrt{\chi^2_\nu/\nu}$, where $z = L\varepsilon$, L is the Cholesky factor of the sample covariance and $\varepsilon \sim N(0, I)$. The scaling keeps the sample covariance while adding fat tails. The degrees of freedom $\nu = 4.22$ come from the portfolio fit above. The portfolio loss is $-w^\top x$, and VaR and ES are read with the historical estimator on the simulated losses (seed 42).

4.6 EWMA and backtest

Each day, VaR is estimated on the previous 250 days only and compared with the next day's return. A breach is a loss larger than the VaR. A fifth model, EWMA, is added: a normal VaR with $\sigma^2 = \sum_k \lambda^k r_{t-k}^2 / \sum_k \lambda^k$ and $\lambda = 0.94$ (the daily RiskMetrics value, J.P. Morgan/Reuters, 1996), applied to the same 250 days. Recent days weigh more, so it reacts faster than the plain normal model.

Test	Question	Statistic
Kupiec (1995)	Is the breach count consistent with 1%?	$LR = -2\ln\dfrac{(1-p)^{n-x}p^x}{(1-\hat{p})^{n-x}\hat{p}^x}$, $\hat{p} = x/n$
Christoffersen	Are breaches independent?	Markov chain of breaches against a constant rate

Both statistics are compared with a χ^2_1 distribution. Here n is the number of days, x the number of breaches and $p = 1\%$. The Christoffersen test compares $\hat{p}_{01} = P(\text{breach} \mid \text{none yesterday})$ and $\hat{p}_{11} = P(\text{breach} \mid \text{breach yesterday})$ with one common rate.

5 Method Comparison

Method	Strength	Weakness
Historical	No distribution assumption	Tail rests on 15 points (3 to 4 in a 250-day window)
Normal	Stable, closed form, two moments	ES 3.40% against 4.77% observed
t fit	Fat tails with one extra parameter	Symmetric; ν unstable between windows
Monte Carlo	Extends to joint and non-linear cases	Equals the t model here; sampling noise
EWMA	Follows volatility within weeks	Normal tails; sensitive to λ

For a linear portfolio, a linear combination of a multivariate t vector is a univariate t with the same ν. The Monte Carlo model therefore gives almost the same answer as the Student-t fit (3.40% against 3.37%), and its value would appear only with options or other non-linear positions. Historical simulation is the benchmark to explain to a committee, the t model is the better statistical estimator on short windows, and EWMA is the right reference for how fast the number should move.

6 Results

Full sample (1,494 days, one-day 99%, in % of portfolio value):

Model	VaR 99%	ES 99%	ES/VaR
Historical	3.37	4.77	1.42
Normal	2.97	3.40	1.15
t fit	3.37	4.63	1.37
Monte Carlo (t)	3.40	4.65	1.37

The normal model gives a lower VaR and ES than the historical one because its single volatility is set mostly by ordinary days, and the excess kurtosis of 6.8 is left out. The gap grows with the confidence level: at 95% the normal VaR (2.10%) is above the historical one (1.80%), at 99.5% it is 0.58 points below (3.28% against 3.86%). The five worst days are all in 2020 except 13 June 2022 (-5.45%); the worst, 12 March 2020, is -11.06%. Without 2020, the historical ES falls from 4.77% to 3.70%. These numbers are in-sample, so they say how the models describe 2020 to 2025, not how good they are as forecasts.

Out-of-sample backtest (1,244 days, 4 January 2021 to 31 December 2025, 12.4 breaches expected):

Model	Breaches	Rate (%)	Kupiec p	Christoffersen p
Historical	19	1.53	0.083	0.032
Normal	20	1.61	0.048	0.039
EWMA	21	1.69	0.026	0.048
t fit	12	0.96	0.900	0.004

- **Count:** the Student-t model matches the expected number of breaches. Normal and EWMA fail the Kupiec test at 5%, and the historical model is close to the line ($p = 0.083$).
- **Clustering:** all four models fail the Christoffersen test. The Student-t model has the right count but the clustering is worst.
- **Risk source:** Ethereum is 10% of capital and 28.3% of portfolio volatility. NVIDIA adds 18.5%. TLT is 20% of capital and 3.2%.

The historical, normal and EWMA models are breached 1.5 to 1.7 times as often as they should be: 19 to 21 breaches against 12.4. All three use the 250 days before the date, and after a calm year the window holds no crash, so the VaR sits near its minimum (1.70% for the historical model) and the next shock breaks it. The Student-t VaR is smoother, between 1.94% and 5.04% against 1.70% to 5.45%, because the fit uses all 250 observations and not the three in the tail. The test does not prove that the t model is right: with 12 breaches it has little power, and the other models are only 7 to 9 breaches away.

The Christoffersen result needs care. Breaches cluster in 2022 (4 to 8 per model) and in April 2025, when the Student-t model is breached on 3, 4 and 7 April. For that model, 12 breaches include 10 that follow a non-breach day and 2 that follow a breach day, so $\hat{p}_{01} = 0.8\%$ and $\hat{p}_{11} = 16.7\%$. The rejection rests on two consecutive pairs. It is a warning about April 2025, not proof that independence fails every month.

Year	Historical	Normal	EWMA	t fit
2021	2	3	5	1
2022	7	8	4	4
2023	1	0	2	0
2024	4	4	4	3
2025	5	5	6	4

At 1% a year holds about 2.5 breaches. Every model is far above that in 2022 and in 2025, and at or below it in 2023. Breaches follow the volatility regime, not the calendar.

Risk contribution (weight times marginal contribution, sums to 100%):

Asset	Weight (%)	Annual vol. (%)	Risk contribution (%)
Ethereum	10	83.5	28.3
NVIDIA	10	53.9	18.5
Apple	10	32.8	11.1
Microsoft	10	30.7	10.8
Airbus	10	40.3	9.9
LVMH	10	30.7	8.8
L'Oréal	10	25.0	6.2
Sanofi	10	23.5	3.2
TLT	20	18.5	3.2

Equal weights are not equal risk. Ethereum, with 83.5% volatility, carries almost three times its capital share. TLT is 20% of capital but 3.2% of risk: its correlation with the other eight assets is between -0.05 and 0.03 and its own volatility is low. That low contribution is a fact of 2020 to 2025 and need not hold in another rate regime. The table says where volatility comes from, not where the tail comes from: contributions to ES can differ.

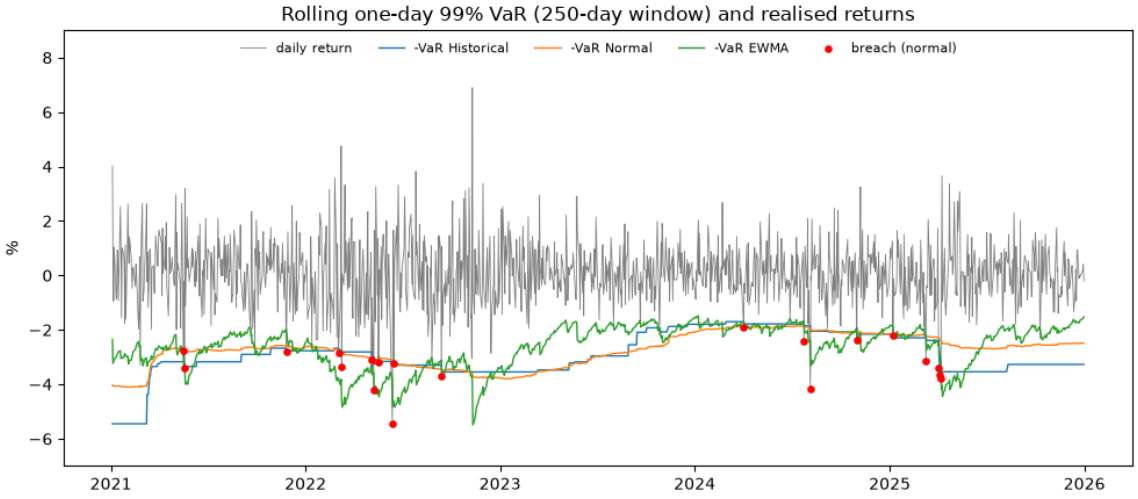


Figure 1: Rolling one-day 99% VaR (250-day window) and realised returns. Red dots are normal-model breaches.

7 Worked Example

Historical. There are $n = 1494$ returns. The 1% quantile sits at position $0.01 \times (n-1) = 14.93$ (counting from 0) in the sorted list, between the 15th worst return (-3.400%) and the 16th (-3.368%). Interpolating gives $q = -3.37\%$, so VaR $= 3.37\%$. Fifteen returns lie at or below q and their average is -4.77%, so ES $= 4.77\%$. On an illustrative EUR 1,000,000 portfolio the VaR is EUR 33,700 and the ES EUR 47,700.

Normal. $\sigma = 1.275\%$, $z = 2.3263$, $\varphi(z) = 0.02665$:

$$\begin{aligned} \text{VaR} &= 2.3263 \times 1.275\% = 2.97\%, \\ \text{ES} &= 1.275\% \times \frac{0.02665}{0.01} = 1.275\% \times 2.665 = 3.40\%. \end{aligned}$$

Student-t. $\nu = 4.224$, $s = 0.926\%$, $q = t_\nu^{-1}(0.99) = 3.639$, $f_\nu(q) = 0.00923$:

$$\begin{aligned} \text{VaR} &= 0.926\% \times 3.639 = 3.37\%, \\ \text{ES} &= 0.926\% \times \frac{0.00923}{0.01} \times \frac{4.224 + 3.639^2}{3.224} = 0.926\% \times 0.923 \times 5.418 = 4.63\%. \end{aligned}$$

Kupiec test for the Student-t model: $n = 1244$, $x = 12$, $p = 0.01$, so the expected count is 12.44.

$$\begin{aligned} \hat{p} &= 12/1244 = 0.00965, \\ LR &= 0.0159 \quad \Rightarrow \quad p\text{-value} = 0.900. \end{aligned}$$

For the normal model, $x = 20$ gives $LR = 3.919$ and a p-value of 0.048, below 5%.

Sensitivity. The first line is illustrative, the others rerun the repo code.

- If σ doubled to 2.55%, the normal VaR and ES would double to 5.93% and 6.80%.
- At the same $\sigma = 1.275\%$ but with $\nu = 3$, 10, 30 the t VaR is 3.34%, 3.15%, 3.03% and the ES 5.15%, 3.84%, 3.53% (the fitted $\nu = 4.22$ gives 3.37% and 4.63%). The VaR moves by 0.3 points, the ES by 1.6. At $\nu = 30$ the model is close to the normal one (2.97% and 3.40%).

- Scaling the normal VaR to ten days by $\sqrt{10}$ gives $2.966\% \times \sqrt{10} = 9.38\%$ (from the unrounded VaR; the rounded 2.97% would give 9.39%), which assumes independent days and zero drift.
- Window length in the backtest (breach counts):

Window	Days tested	Expected	Historical	Normal	EWMA	t fit
125	1,369	13.7	23	19	23	15
250	1,244	12.4	19	20	21	12
500	994	9.9	12	14	16	10

With 500 days no model fails the Kupiec test at 5% (the smallest p is 0.076, for EWMA); with 125 days the historical and EWMA models fail ($p = 0.021$ each). The test periods differ (a longer window starts later and drops 2021), so the three rows are not strictly comparable. Still, the pass or fail verdict depends on a window choice, which is a reason to report the breach counts next to the verdict. All four models fail the Christoffersen test at 500 days ($p \leq 0.024$).

8 Limits and Extensions

- **Short sample:** 2020 contains the March crash, which drives the historical ES. The sample is six years, so the 1% tail rests on about 15 observations.
- **Fixed weights:** daily rebalancing to fixed weights is assumed, with no transaction costs, and the portfolio is not a real book.
- **One-day horizon:** a zero mean is standard at one day but does not hold for longer horizons, and a regulatory number over ten days needs a proper model for that horizon.
- **Test power:** with about 12 expected breaches, Kupiec and Christoffersen have limited power. The results are model outputs, not recommendations.
- **ES is not backtested:** the tests check the VaR only. Nothing here checks whether the ES is right.

Three extensions would address these limits. Filtered historical simulation (rescale past returns by a GARCH or EWMA volatility, then take the quantile) would let the historical model follow volatility and should reduce the clustering. A skewed t, or a generalised Pareto fit to the worst returns, would model the left tail on its own. An ES backtest, together with the Basel traffic-light zones (Basel Committee, 1996), would test the number that matters for capital.

9 Conclusion

On this portfolio the normal model understates the loss on a bad day by more than a quarter (ES 3.40% against 4.77%), and the models that look back 250 days at a plain quantile or volatility are too low after calm periods. A desk should therefore watch the breach count and the clustering together, and treat any one VaR figure as the output of a window and a distribution, never as a fact.

References

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